

Developer Report — Fluid Flow & Heat Transfer Engineering Suite

PE258/PE262 Capstone Project

1. An Interesting Engineering Insight the App Reveals

Using the Pipe Flow Analyser's pressure-drop-vs-flow-rate chart, it becomes immediately visible that pressure drop does not scale linearly with flow rate once the flow transitions from laminar to turbulent: in the laminar regime ($f = 64/Re$) the pressure drop grows roughly with the first power of velocity, but once Re exceeds ~2300 and the Swamee-Jain turbulent correlation takes over, pressure drop grows approximately with velocity squared. Sweeping the flow rate slider across the laminar-turbulent transition on the interactive chart shows a visible change in curvature at that point — a direct, hands-on illustration of why pipeline engineers try to avoid operating right at the transition, since small changes in flow rate there cause disproportionately large changes in pumping pressure requirements.

2. A Technical Challenge Overcome

The Darcy friction factor for turbulent flow is defined implicitly by the Colebrook equation, which has no closed-form solution and normally requires an iterative (e.g. Newton-Raphson) solve. Implementing an iterative solver inside a Streamlit app that recalculates on every slider movement risked either slow responsiveness or convergence failures for edge-case inputs (e.g. very low roughness or Reynolds numbers near the laminar-turbulent boundary). This was resolved by replacing the iterative Colebrook solve with the explicit Swamee-Jain approximation, which is accurate to within about 1% of Colebrook across the correlation's valid range and evaluates in a single closed-form expression — eliminating both the responsiveness concern and any risk of non-convergence, while remaining engineering-grade accurate. This substitution was verified by manually comparing friction factors from both methods at several Reynolds numbers before adopting it app-wide.

3. One Thing to Add With More Time

Given more time, the next addition would be a multi-layer composite wall option for the Heat Transfer Calculator's conduction tab (generalising the current single-layer `PlaneWallConduction` class to sum thermal resistances in series across several layers, including convective boundary layers on both sides) — this would let the same module handle realistic building-envelope or insulated-pipe problems rather than only a single homogeneous slab, and would pair naturally with a resistance-breakdown table similar to the one used in the earlier composite-wall thermodynamics assignment.

Repository: [PASTE GITHUB URL] | Live app: [PASTE STREAMLIT URL]